RHESSys in Practice:

Applied Ecohydrological Modeling

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Welcome to the RHESSys Manual

Authors

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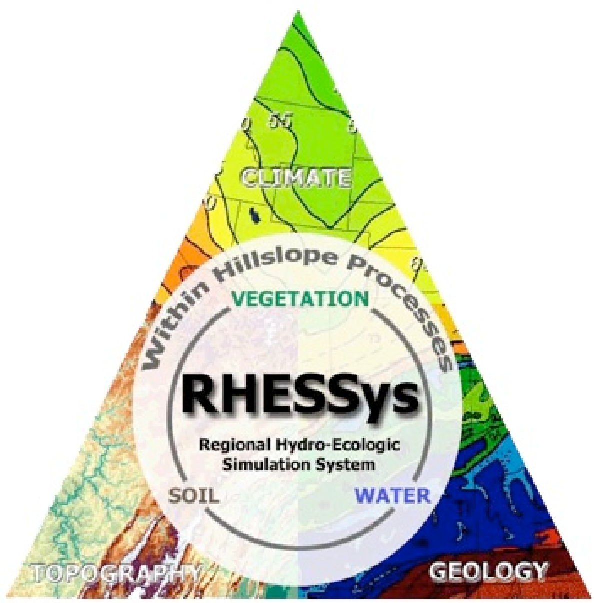

Purpose

Include info about the manual, why it exists, what it is intended for

Scope

- What to expect from this book (user will learn….)
- What the manual covers (practical step-by-step guidance)
- What it does not cover (underlying theory of the model, all possible functionality)

Intended Audience

- Who this is for/Who should be reading this

Assumed basic prerequisite knowledge

- **General Modeling**
- **Basic R/Rstudio**
- **GIS and Raster/Vector**
- **Basic Unix/Linux environment (shell, command line, text editor)**
- **Git/GitHub**

About this Manual

Conventions

- How to read this document
- Key/Legend for document formatting - regarding use of icons, symbols, different font indications

Manual Structure

- organization
- navigation
- links to sections
- potentially a link to a quick start guide

Support/Resources

- Potentially include author info contact info (?)
- links to
  - R/Rstudio resources
  - Git/Github
  - RHESSys GitHub repositories (RHESSys code, RHESSysPreProcessing, RHESSysIOinR, ParameterLibrary)
  - RHESSys Wiki(s)
  - (Potentially) a FAQ page, a user community forum (i.e. 'issues' section on RHESSys Github repo?)
  - other relevant resources

Part I

Getting Started

Part II

- This part is about understanding the logic of this tool and setting the computational environment
- intro to what is covered in the section
- expected outcome for this section: functioning modeling environment and clear strategic plan

Part III

Chapters in this section

Introduction

Understanding what RHESSys is

Installation

Establishing the modeling environment

Project Design

Planning your project strategy

Example Dataset

Explore the example data set

1 Introduction (The Model)

1.1 Overview of RHESSys

1.1.1 What RHESSys is

- Description

1.1.2 Capabilities

- Examples of what can be done with the model

1.1.3 Strengths

- Integrated Ecohydological Modeling
- Spatial distribution of the landscape
- Feedback loops
- Fire model

1.1.4 Limitations

- High data demands
- Can be computationally intensive
- Maybe any processes that are not well represented? Let users know if the model won't fit their research question

1.1.5 Comparison of RHESSys to other models

- what this model is appropriate for
- what the trade-offs are

1.2 Recommended reading/Publications

1.2.1 Foundational papers

- publication links with brief descriptions

1.2.2 Overview/Review papers

- publication links with brief descriptions

1.2.3 Applications

- publication links with brief descriptions

1.3 Workflow overview

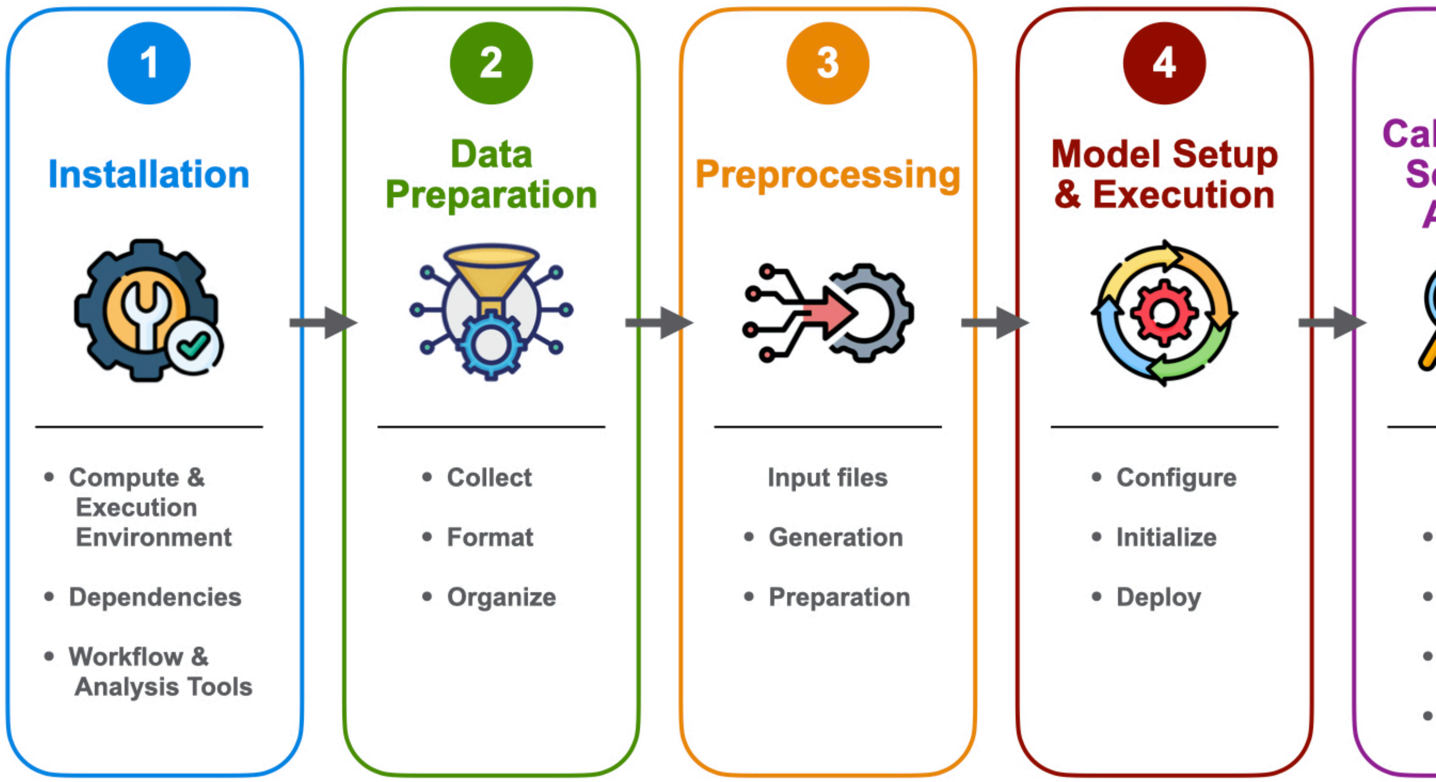

Figure 1.1: **RHESSys Modeling Workflow Overview**

- Step 1: Installation - Set up the RHESSys environment
- Step 2: Data Preparation - Gather and format spatial, climate, and validation data
- Step 3: Preprocessing - Generate the landscape representation (Worldfile) and routing (Flowtable) input files
- Step 4: Model Setup and Execution - Define the simulation rules and run RHESSys
- Step 5: Calibration & Sensitivity Analysis - Condition the model (Initialize/reach equilibrium), refine parameters, validate performance
- Step 6: Post Processing - wrangle, analyze, and visualize output data

1.3.1 Key concepts and terminology

(define frequently used terms)

- Spatial hierarchy (basin, hillslope, zone, patch, stratum)
- Input File types (Worldfile, Header, Flowtable, Tecfile, Parameter/Climate files, Output filters)
- State variables
- Spatial/Temporal step (i.e., basin daily, patch monthly)

2 Installation (The Software)

2.1 Setup options

(description of each, who should use, when to use, recommendation, link to corresponding install instructions)

2.1.1 Local Installation (macos, Linux)

- more advanced users
- users who want more flexibility

2.1.2 Containerized setup (Docker)

- beginners
- Windows users
- portability

2.1.3 HPC/Cloud (SIF/Singularity)

- high-performance computing environments
- cluster users
- large scale, high resolution simulations

2.2 RHESSys Installation Steps

(link to instructional video or videos if each option should be separate)

2.2.1 MacOS

(include section for Linux as well if necessary or where differs)

Code Dependencies and Operational Requirements

- Terminal application
- Xcode Command Line Tools (compilers/build tools)
- Privileges (i.e. Administrator access)
- Homebrew
- Git
- Libraries (i.e. NetCDF)

Installation steps

- Download code from GitHub
  - Zip file
  - Clone repository (main, develop)
- Compile
- Shell configuration (adding to path)
- Verify installation
- Common issue and solutions

2.2.2 Docker

- Install Docker Desktop
- Obtaining RHESSys image from the Docker Hub

2.2.3 SIF

- Obtaining RHESSys .sif image

2.3 Installing Workflow & Analysis Tools

(link to instructional video)

2.3.1 R Ecosystem

- R (The engine)
- RStudio (The interface)
- Standard Packages (Extensions - Libraries that expand functionality)

2.3.2 WhiteboxTools for Geospatial Processing

2.3.3 RHESSys-Specific R packages

- RHESSysPreprocessing
- RHESSysIOinR

3 Project Design & Planning (The Strategy)

- Why project design matters - what the model is being built to accomplish
- Project design decision should be made before gathering/preparing data and constructing the model
- The cost of poor planning
- Model design is a sequence of interconnected decision

3.1 Model Purpose (The Anchor)

- Establish the scientific objectives of the project, is the purpose to:
  - gain system understanding through exploration
  - predict future conditions
  - compare management scenarios
- Define clear research questions that the model must address
- Identify needed outputs (e.g., hydrology, vegetation dynamics, carbon fluxes, fire behavior, scenario comparisons)
- Use research questions to constrain:
  - spatial scale
  - temporal scale
  - model complexity
- Ensure alignment between conceptual goals and RHESSys capabilities

3.2 Modeling Extent (The Frame)

- Define the spatial extent of the modeling domain, the area of interest:
  - patch-scale
  - hillslope-scale
  - watershed/basin-scale
- Match spatial extent to research questions and process scale
- Ensure the domain captures all relevant hydrologic/ecologic processes
- Restrict the model's scope to what is needed for analysis
- Establish initial spatial frame before deciding resolution or structure

3.3 Spatial Resolution vs. Computational cost (The Trade-off)

- Determine appropriate spatial resolution for the model domain
- Understand trade-offs between:
  - fine resolution = higher realism, higher computational cost
  - coarse resolution = faster runtime, reduced spatial detail
- Consider constraints:
  - available computing resources
  - simulation time requirements
  - data limits (DEM resolution, input data available)
- Ensure resolution is consistent with:
  - scale of the research questions (i.e. patch, subbasin, watershed)
  - input datasets
  - practical limits of available computing infrastructure

3.4 Spatial Landscape Partitioning (The Structural Design)

Multi-scale routing (MSR) allows for fine scale interactions within the smallest fundamental landscape unit. While this section briefly introduces MSR as a way of thinking about patch organization, its full implementation is deferred to Chapter 18 in the Advanced Applications section, where the methodology is discussed in full.

- Break landscape into RHESSys fundamental units (patches)
- Organize patches into patch families to represent spatial structure
- Represent land cover composition within patch families:
  - homogeneous landscapes = simpler structure (can be a single patch in a single patch family)
  - heterogeneous landscapes = more complex partitioning to capture variability in land cover or management regimes (may be multiple/many patches per family)
- Determine number and configuration of patches based on:
  - vegetation diversity
  - landcover/soil variability
  - management objectives
- Consider how patch family composition will support model scenarios:
  - design for scenario comparability (consistent model structure so differences between simulations only reflect the intended experimental change)
  - ability to pre-structure the landscape for management experiments
  - flexible scenario implementation without redesigning the landscape configuration
- may include a scenario-based design example to illustrate the utility

3.5 Data Availability, Synchronization & Temporal extent (The Feasibility)

- Availability of input data resources that support project objectives
- Climate data continuity, consistent time periods across all inputs
- Simulation period supported by temporal data extent
- Availability of observational data for evaluating model performance

3.6 Functional constraints (The Capability)

- ability of model setup to accomplish project goals
- available model processes
- ability of model to output variables needed to answer project questions
- availability of feature specific inputs to use specific process modules - i.e. wind speed/direction & relative humidity data to run the fire model

3.7 Summary: Building a Project Strategy

- Project design advances through a structured progression of limitations:
  - research questions → define purpose
  - purpose → determines spatial extent
  - extent → constrain resolution
  - resolution → informs spatial partitioning
  - partitioning → depends on data availability
  - data availability → determines functional feasibility
- Each step progressively narrows design choices
- Ensures model is both: ->
  - aligned with the projects goals
  - practical to build and run with the available data and resources ->

4 Example Dataset (The Sandbox)

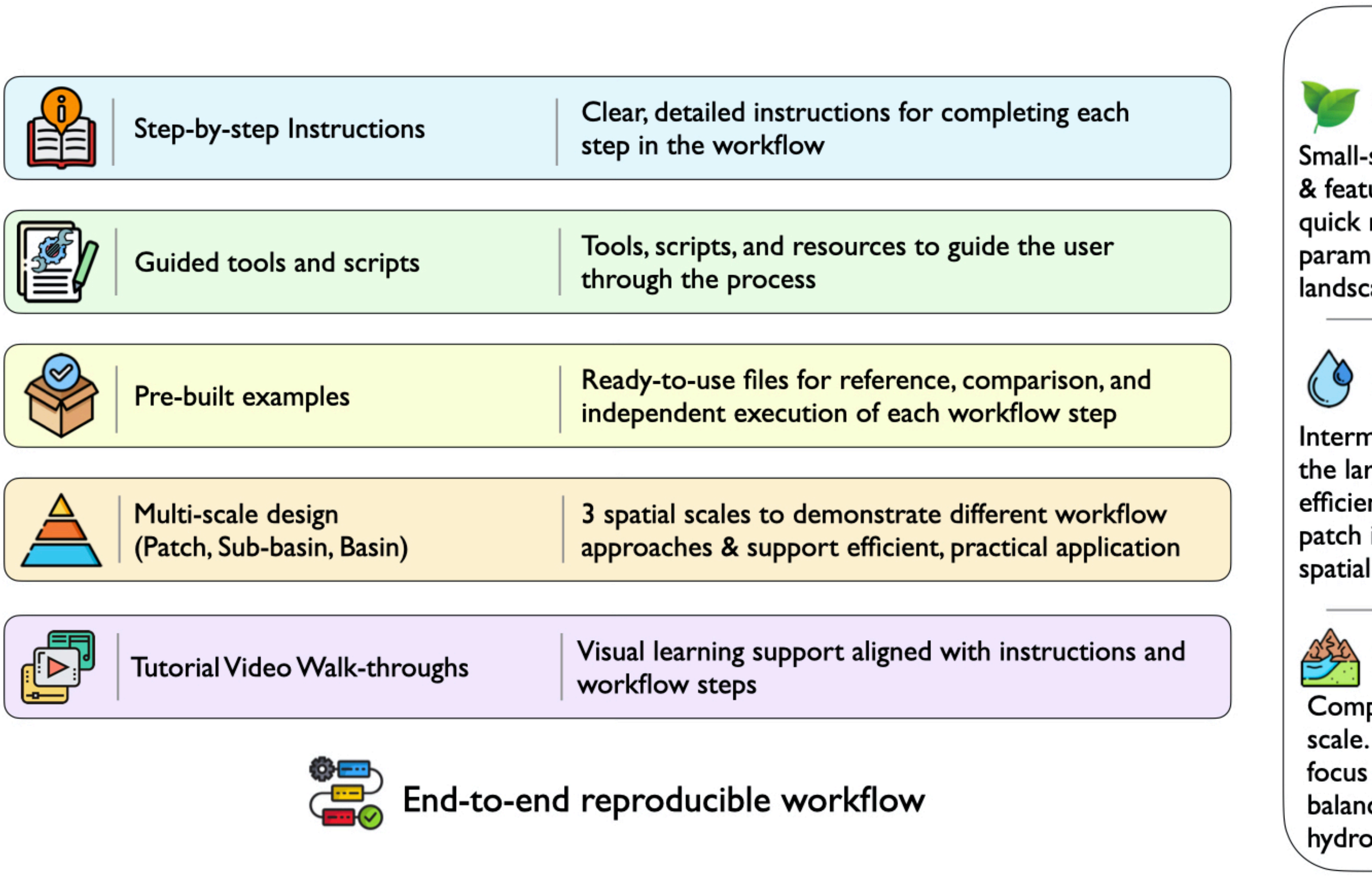

Figure 4.1: **Instructional System Design**

4.1 Study Site Overview

- Location → Sagehen
- Characteristics

4.2 Dataset Contents

- Standard project directory structure

- Spatial files
- Climate Base station examples
  - for local data (point, station)
  - for gridded products
    * ascii formatted
    * netcdf
- Meterological files
  - Precipitation
  - Temperature
  - Wind
  - Relative Humidity
  - Formatted for:
    * local data (point, station data)
    * gridded products
- Definition (default) Files
- observational data
  - streamflow timeseries
  - reference values (i.e. LAI, Height, Biomass)
- Preprocessing Rscript
- Worldfile and Flowtable examples (for a patch, a sub-basin (hillslope), and a full basin)
- Worldfile Header example
- Tecfile examples
- Output Filter examples
- Simulation Rscripts
  - will include functions to generate RHESSys specific supporting files
  - single run
  - multi-run
  - spin-up
  - sensitivity analysis
  - calibration
- Post-Processing Rscripts
- Example model output results
- Dataset Download instructions/link

4.3 How to use this data

- Contextual understanding
- Performance Evaluation - verification for user generated files
- Troubleshooting
- Modular learning - start at any step in the manual

Part IV

From Data to Simulation

- This part is about building the model, constructing the foundation for a fuctional simulation
- intro to what is covered in the section
- expected outcome for this section: creating input in order to run the model, generate output, and interpret results

Chapters in this section

User-Aquired Data

Gathering and formatting the necessary ingredients

RHESSys Specific Input Files

Understanding the core file anatomy

Preprocessing

Assembling the model foundation

Implementation & Execution

Launching the simulation

Postprocessing

Gaining model result insight

5 User-Acquired Data (The Ingredients)

- here introduce what each item is, why necessary, sources to obtain

5.1 Spatial Data

- Required
  - DEM (source to obtain)
    - DEM Derivatives - generation discussed in later section:
      - Watershed Basin boundary
      - Sub-basins (hillslopes)
      - Slope/Aspect
      - E/W Horizons
      - Stream Network
      - Patches
  - Watershed outlet - gauge location coordinates
- Recommended (typical) or when applicable:
  - Vegetation cover
  - Soils
  - Landcover/Landuse
  - Roads
  - Impervious areas
  - Climate base stations
  - Aspatial rules ID map for multi-scale routing

5.2 Climate Data

- Timeseries data
- Required meteorologic data
- Optional meteorologic data
- About Supported options
  - local station data (point)
  - gridded products
    - ascii
    - NetCDF
  - pros/cons of using each
  - sources to obtain

5.3 Observational Data

- used to constrain and evaluate hydrologic and ecophysiological parameterization
- Types of observed data
  - utility (how will be used, i.e. veg values to initialize parameters, also to validate performance)
  - Hydrologic (i.e., Streamflow, Snow, Soil moisture)
  - Vegetation (i.e, LAI, Height, NPP, ET, rooting depth)
  - Soil (type, properties, texture) (polaris)
  - sources to obtain

6 RHESSys-specific Files (The Toolkit)

RHESSys relies on a collection of specialized files that define model parameters, climate inputs, spatial structures, routing relationships, execution schedules, and output behavior. Together, these files form the model's anatomy. Among them, the Worldfile occupies a central role and is treated separately in Chapter 7 due to its complexity and importance. The files presented in this chapter provide the parameters, forcing data, and control information that are associated with the spatial units defined in the Worldfile.

6.1 Definition (Default) Files: The Library

- describe typical meteorological, soil, vegetation, and land use characteristics
- File types associated with each level
- Parameter values constant through simulation

6.2 Base station file: The Climate Hub

- points to the set of climate data input files
- provides climate station identifier
- connects climate forcing to zones in the worldfile

6.3 Header File: The Directory

- an index of all the definition/climate files RHESSys needs to run a simulation
- connects the Worldfile to the Climate/Definiton files - how the ID's get linked

6.4 Flowtable: The Plumbing

- Architecture
- Topographic data about each patch
- Routing, Patch connectivity, neighbors
- Partitioning flow, upland patches, stream patches

6.5 Tecfile and Output Filters: Simulation Control

6.5.1 Tecfile: The Alarm Clock

- Manage timing of output printing
- Control simulation events (schedule when non-continuous events occur, i.e. fire, land use change)
- Trigger a worldfile redefinition to update properties of the landscape

6.5.2 Output filters: The Sieve

- specify criteria for selecting and filtering the output data
- Filter rules
- Filter conditions
- Filter Parts: timestep, spatial level, output format/name
- Variables associated with Spatial/Temporal leval of output
  - sub-level output
  - sub-routine output
- Variable reassignment: name reassignment, arithmetic expressions

6.6 The Operational Framework

The files described in this chapter provide the information and controls required to configure a RHESSys simulation. While each file serves a distinct purpose, together they operate through a shared set of spatial identifiers that link model inputs to discrete spatial units within the watershed.

These spatial units provide the framework for mapping the model's input data. Each parameter and climate forcing input ultimately depends on this spatial structure to determine where and how it is applied during simulation.

However, the definition of this spatial structure itself has not yet been described. The files in this chapter assume the existence of a hierarchical representation of the watershed, but do not define how that structure is constructed or organized.

The next chapter addresses this missing component directly by introducing the Worldfile—the hierarchical spatial framework that defines the structure of the modeled landscape and provides the foundation upon which all files described in this chapter operate.

- New figure showing connections among files

7 Worldfile (The Anatomy)

Chapter 6 described how RHESSys-specific files assign parameters, climate forcing, routing rules, and simulation controls to spatial units within a watershed. Those assignments depend on a predefined spatial hierarchy, but that hierarchy has not yet been explained.

The Worldfile provides this structure. It defines the hierarchical organization of the watershed into discreet spatial units, establishing the spatial framework to which model inputs are applied.

This chapter examines the architecture of the Worldfile, the identifiers that connect it to supporting files, and the spatial framework that underpins every RHESSys simulation.

- figure here to illustrate the worldfile and how input files are linked to the landscape
- refer to throughout the chapter

7.1 Spatial Hierarchy: The Landscape Representation

- Hierarchical spatial containment structure
  - Basin
  - Hillslope (sub-basin)
  - Zone
  - Patch Families containing Asptatial patches
  - Stratum
- Containment rules
- processes associated with each level

7.2 Structural Index: Worldfile Architecture

- worldfile format
- spatial unit ID's

- parent-child linking logic (stratum/patch ID's built on patch family ID)
- values for stores and fluxes organized at each spatial level
- Linking identifiers (ID's) pointing to the associated definition & climate files

7.3 Definition File Link: Parameter Integration

- def files define WHAT parameters applied to associated level, Worldfile defines WHERE they live spatially

7.4 Climate Link: Base Station Integration

- how climate maps to spatial units
- how climate inputs are distributed across the landscape
- spatial resolution of climate inputs - the spatial unit applied to

7.5 Template File: The Recipe

- set of instructions used to build the model's digital spatial landscape
- links GIS raster data to the hierarchical structure
- maps rules from input files to spatial structure
- contents:
  - Spatial world objects
  - State variables
  - Default and base station ID's
- program applies the template instructions to each spatial unit, expanding the rules to the extent of the defined watershed area, resulting in the Worldfile

7.6 Integrated Spatial System

- Worldfile is the integration point for the Chapter 6 inputs
- spatial manifestation of input information

8 Preprocessing (The Assembly)

- the how to section
- instructions to process/generate files
- how to format data
- link to video(s)

8.1 Environmental Setup

- Standard directory structure
- naming convention

8.2 Raster Processing

- Spatial processing using WhiteboxTools and R

8.3 Climate Fles

- gridded product formatting
  - ascii format
  - netcdf format
- local station data formatting
- base station

8.4 Definition Files

- Format - required (ID) and optional parameters
- Base Files/Default values
- Parameter file library
- some parameters may need expanded information, i.e. phenolog flag: static, dynamic, drought

8.5 Generating the Worldfile and Flowtable

8.5.1 The Template File

- Format and Initialization
  - World objects
  - State variables
  - Default and base station ID's
- Required and Optional state variables
- Editing the template, setting initial conditions

8.5.2 Aspatial rules file for Multi-scale routing

- Format
- Set up for single rule
- Set up for multiple rules - requires map

8.5.3 RHESSysPreprocessing R package

- About
- Prerequisites: spatial data, template, asp rules file
- R markdown workflow/script - functions
- Run the function to generate Worldfile/Flowtable

9 Implementation & Execution (The Launch)

9.1 Simulation Prerequisites

- RHESSys environment already set up (i.e. Docker image, compiled version of RHESSys, SIF file)
- Worldfile
- Flowtable
- Formatted climate files/Base station
- Set of definition files corresponding to the project

9.2 The RHESSysIOinR environment

- Bridge between RHESSys code and the R environment
- Package overview

9.3 RHESSysIOin R Script configuration

- Rmd script description
- input mapping options
  - temporal parameters (setting start/end dates for simulation & tec events)
  - referencing input files (worldfile/flowtable)
  - directing where output is written
  - command line options
- generating the header file
- generating the tecfile

9.4 Data output filtering

- Building filters in R
  - Combine multiple filters into single R object
- Output filter text file outside of R

9.5 Running the model

- Single run
- Multiple runs

9.6 Example Dataset

- Files to refer to

10 Post-processing (The Insight)

Post-processing is the stage where raw model outputs are transformed into something interpretable, comparable, and useful. This chapter includes example datasets, scripts, and figures that demonstrate a typical post-processing workflow from raw output to final interpretation.

- link to how-to video(s)

10.1 Output Data Wrangling

- Reading output files into R
- Combining outputs across runs or scenarios
- Reshaping and organizing data
- Calculating derived variables and summaries
- Preparing datasets for analysis and visualization

10.2 Output Interpretation

- Exploring patterns and trends
- Comparing scenarios
- Assessing model behavior
- Identifying unexpected results
- Connecting outputs to research questions

10.3 Output Visualization

- Time-series plots
- Generating/viewing Spatial data
- Comparative figures
- Summary statistics and tables
- Communicating uncertainty

10.4 Output Synthesis

- Integrating results across spatial, temporal, process scales
- Summarizing key findings
- Interpreting results to inform decisions or next steps
- Relating results back to model objectives

10.5 Example Dataset

- Example output files
- Example R post-processing scripts
- Example visualizations
- Example Files and scripts referenced throughout the chapter

Part V

Refinement and Evaluation

- This part is about moving from just a running model to a credible scientific tool.
- intro to what is covered in the section
  - the iterative cycle - cyclical refinement loop
  - include a figure/diagram/flowchart about the feedback loop
  - refinement not linear path - changing one parameter may require re-balancing the system
  - each phase informs the next - often requiring return to previous step
- expected outcome for this section: learning the process to ensure a balanced system

Chapters in this section

Initialization

Achieving model equilibrium

Sensitivity Analysis

Identifying influencing parameters

Calibration

Fine-tuning parameters

Validation

Evaluating parameter performance

11 Initialization (The Equilibrium)

Introduction to:

- Description and purpose of initialization
- State variables that require initialization
- Initial values can be estimated from literature, observations, default parameter sets
- Spin-up allows model state to adjust from initial estimates
- Different variables may need different lengths of time to stabilize
- Repeated climate is often used in the spin-up process
- Criteria to determine stabilization
- Established stable conditions used for subsequent simulations
- link to video(s)

11.1 Spin-up/Initializing approaches

11.1.1 Cold start

- starting with very low/zero soil/veg/litter C/N values

11.1.2 Warm start

- reduces model spin-up time
- initialize with literature values or estimates from previous simulation

11.2 Spin-up Strategy

- using a single patch or hillslope
- transfer values to larger watershed
- static/dynamic spin-up

11.3 Spin-up Procedure

- Assign initial state variable values
- Include a spin-up tecfile event
- Establish time period for spin-up
- Standard spin-up output variables for evaluation
- Conduct spin-up simulation
- State-of-the world output Worldfile generated at end of spin-up period

11.4 Evaluate convergence

- Assess whether key state variables have stabilized/reached target value
- Continue spin-up if substantial trends remain
- Use output worldfile from previous simulation as starting condition for next simulation
- Determine baseline equilibrium
- Stable conditions stored in final spin-up state-of-the-world Worldfile as starting point future simulations

12 Sensitivity Analysis (The Influence)

This step relies on a set of R-based scripts and tools that manage the workflow from start to finish. These are tied directly into example datasets so the process can be reproduced and tested easily.

12.1 Tools and Implementation Framework

- R scripts and workflow tools for running sensitivity analysis
  - reference to example data set, explanation of workflow process and functions
- Formalize and standardize workflow for inclusion in RHESSysIOinR
  - establish consistent interfaces between model runs and analysis scripts

12.2 Defining the Parameter Space

- Establishing realistic parameter bounds and constraints
- Using literature values and expert knowledge to inform ranges
- Sampling strategies to explore the parameter space (e.g., random, Latin hypercube)
- Ensuring parameter space/parameter combinations are ecologically plausible

12.3 Running Simulation Experiments

- Automating model runs across parameter sets using R functions/tools
- Managing batch simulations and computational efficiency
- Structuring iterative runs for scalable exploration of uncertainty

12.4 Identifying Influential Parameters

- Evaluating how individual parameters influence model outputs
- Detecting parameter interactions and non-linear effects
- Linking parameter sensitivity to underlying model processes
- Comparing sensitivity across different output variables or metrics

12.5 Parameter Selection and Interpretation

- Defining target outputs and evaluation criteria
- Quantifying uncertainty and variability in model response
- Using statistical and machine learning tools (e.g., random forests, variance-based methods) to estimate parameter importance
- Ranking parameters performance by influence and stability across simulations
- Interpreting results in terms of model behavior and process understanding

12.6 Reporting & Diagnostics

- Visualizing sensitivity results
- Summarizing key drivers of model behavior
- Documenting robust vs unstable parameters
- Feeding results back into model for refinement

13 Calibration (The Optimization)

- R scripts and supporting tools used for calibration
- Reference datasets used for benchmarking and testing
- link to video

13.1 Calibration Strategy

- Type of calibration approach (manual, automated, hybrid)
- Individual vs. simultaneous calibration of parameters
- Single-objective vs. multi-objective calibration
- What is held static vs. what is allowed to vary
- Temporal period for calibration (daily, moving window, seasonal period)

13.2 Parameter Selection & Constraints

- Which parameters are being calibrated and why
- Parameter bounds and constraints (physically realistic ranges)
- Fixed parameters and justification for fixing them (i.e. obtained from Sensitivity Analysis)

13.3 Optimization Setup

- Target data used (observed real world data)
- Optimization methods used (i.e. monte carlo)
- Stopping criteria (when to consider calibration is done)
- Number of runs or iterations

13.4 Evaluation

- Objective functions to measure model performance
- Metrics used to compare simulations to observations
- Challenges of equifinality (multiple parameter sets producing similar results)

13.5 Key Challenges

- Equifinality (different parameter sets producing similar results)
- Uncertainty in observations can affect calibration results
- Parameter uncertainty and sensitivity
- Implications for interpretation of calibration results

14 Validation (The Confirmation)”

14.1 Purpose of validation

- Why validate
- How it differs from calibration (no parameter tuning, independent testing)
- What “good performance” means in this context

14.2 Validation data

- Independent observed dataset not used in calibration
- Selecting validation period
- Important differences between calibration and validation datasets (time period, location, conditions)

14.3 Validation setup

- Model configuration carried over from calibration
- Final calibrated parameters used without modification
- Different simulation period used for validation runs

14.4 Performance evaluation

- Metrics used to assess model performance (same or different from calibration)
- Time-series comparisons between observed and simulated outputs
- Graphical evaluation (hydrographs, seasonal cycles, spatial patterns, etc.)
- Summary statistics

14.5 Generalization Performance

- How well the model transfers beyond the calibration conditions
- Performance under different climates, years, or watershed conditions
- Whether performance degrades, improves, or stays stable

14.6 Interpretation

- What the validation results mean for model credibility
- Whether the model is considered reliable for intended use
- Key strengths and weaknesses revealed by validation

Part VI

Advanced Applications

- This part is about exploring complex eco-hydrological scenarios
- intro to what is covered in the section
- expected outcome for this section: ability to move from model development to high impact scientific inquiry

Chapters in this section

Advanced Features

Expanding beyond the baseline

Fire Disturbance

Simulating the spark and igniting the inferno

Disturbance:Vegetation Dynamics

Transforming an evolving landscape

Multi-scale routing

Creating a mosaic landscape

Target-driven spin-up

Aligning with observed ecological states

Parallelization

Accelerating model simulations

15 Advanced Features (The Expansion)

- beyond the baseline
- moving into specialized simulations
- roadmap of why and when to use these advanced tools
- disturbance modeling in RHESSys: altering the carbon cycles and redirecting the flow of water and nutrients from the baseline
- predicting ecosystem resilience and recovery over time

16 Disturbance: Fire (The Inferno)

16.1 Fire Spread

16.1.1 About

- description
- reference key papers

16.1.2 Preparation

- compiling RHESSys-WMFire: Boost/WMFire library
- Prerequisites to running the fire model:
  - Wind speed and direction data
  - Fire definition file parameters
    * calculating the fire return interval
    * calculating wind parameters (EstimateWindDistrExample.R script)
  - Worldfile header edits
  - Add worldfile state variable
  - Command line option

16.1.3 Outputs

- Fire definition parameter options to generate output
  - Fire Sizes output
  - Grid output
- spatial level fire outputs

16.1.4 Fire simulation stragey

- stochastic
- prescribed

16.2 Fire Effects

- about
- Fire effects parameters in vegetation and soil(patch) definition files
- specific fire effects output files

17 Landcover Change (The Evolution)

17.1 Types of Change

- Land Cover/Vegetation change
- biomass pool changes

17.2 Changing the vegetation cover

- Homogeneous changes
- Heteorogeneous changes
- Partial redefinition (thinning)

17.3 Generate a Redefined worldfile

- redefine with an awk script
- redefine with a modified worldfile (redefine worldfile)
  - redefine template
- short simulation to output a changed worldfile
- new redefined worldfile output for use in subsequent simulations

17.4 Applying the Change

- worldfile date extension
- Tecfile date
- Tecfile events
  - redefine_world
  - redefine_world_multiplier
  - redefine_world_thin_remain
  - redefine_world_thin_harvest
  - redefine_world_thin_snags

18 Fine-scale heterogeneity (The Mosaic)

- link to video

18.1 Advantages

- Accounts for forest density reduction, species interactions, and urban eco-hydrology
- Allows for finer scale lateral water and nutrient transport
- Allows for within patch family) transfer of water between neighboring patches

18.1.1 How It Works

Multiscale routing (MSR) allows for interactions (notably the routing of water and shading between) among model units (a patch family unit containing multiple patches) and their associated vegetation types that are typically at scales too small (<30m) to characterize explicitly. Hypothetically, this allows for the (within patch family) transfer of water between neighboring patches of thinned and unthinned trees, or from a sidewalk to a lawn – none of which would typically be represented/accounted for when modeling at a 30m grid (patch) cell.

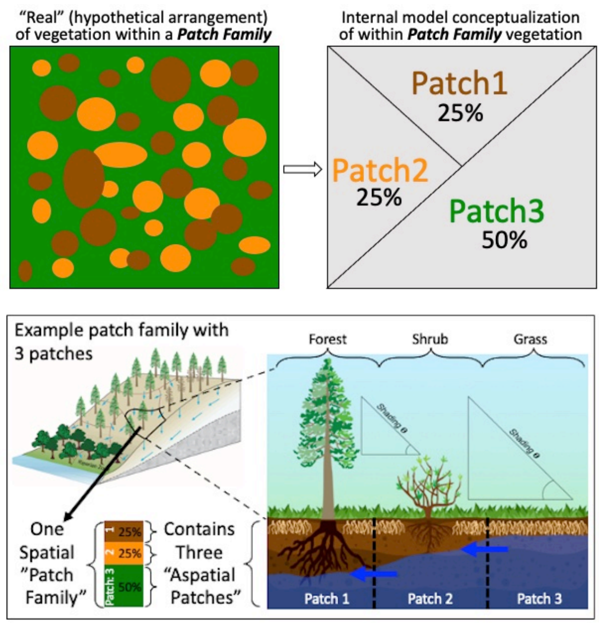

Figure 18.1: **Multi-scale routing**

18.2 Characterization in RHESSys

RHESSys uses patches as the smallest spatial unit. Multiscale routing leverages the existing spatial units in RHESSys by placing multiple patches in the exact same location. These co-located patches together are a patch family, and the routing of multiscale routing, termed "local routing" occurs between these patches, in addition to the normal topographic hillslope routing present in RHESSys which functions as normal.

- Patches reflect distributions of land cover (species, open-space gaps)
- Assumes that these land covers are well mixed within a patch family
- Exchanges based on rules (e.g., root accessibility) and moisture gradients
- Multiple aspatial patches within a family allow for within patch heterogeneity
- Aspatial patches can differ by:
  - vegetation type (biomass, height, root depth)
  - Surface coverage (open areas, thinned areas, remaining trees)
  - surface properties (litter)
- Exchanges of water between trees, between trees and open space (root-based access to neighboring tree stores)

18.3 Multiscale Routing Logic in RHESSys

Multiscale routing can hypothetically simulate any number of water transferring processes that occur at scales smaller than RHESSys models topographically. The first demonstration though will simulate the access/lack of access that trees and other vegetation have due to their roots. New uses of multiscale routing will require rethinking the logic behind how water is routed within a patch family.

18.4 Multiscale Routing in RHESSys – Shading

Shading can occur when the vegetation of other patches within a patch family creates a higher angle to the horizon than the horizon angle map input to RHESSys.

18.5 Multiscale Routing in RHESSys – Thinning and Fire

18.5.1 Thinning works

- Area can't change
- Percent cover changes
- Understory removal happens everywhere
- Selection thinning on specific aspatial patches ### Fire Spread

- Slope and wind are the same across patch family
- Litter load and 1-AET/PET calculated at patch level, turned into probabilities, spatially averaged across patch family ### Fire Effects
- Done per patch

18.6 Set-up

- MSR input files
  - aspatial rules file
  - aspatial rules map also needed if multiple rules are being used
  - Format
- Add Template state variable
- Generating msr worldfile
  - add argument to use aspatial rules file to RHESSysPreprocessing

18.7 Running RHESSys with msr

- Multiscale routing is automatically triggered when the structure of the worldfile & flowtable are formatted for MSR/aspatial patches (which is generated when the asprules command is included when creating the worldfile/flowtable)

18.8 Output

- Output of multiple patches that belong to a patch family
- Patch Family ID in addition to Patch ID

19 Target-driven spin-up (The Alignment)

This approach (target driven spin-up) merges the two existing methods for initializing vegetation carbon and nitrogen stores, where one or more real vegetation parameters derived from a spatial data layer (e.g. LAI from TM or MODIS, stem wood or total biomass from LiDAR, or even stand age maps etc.) are used to define targets for each patch. The model is then run in spin-up mode - which tracks C & N stores for each patch separately until the target value is reached. Once all patches have reached their target, the model outputs a world state that includes the appropriate C and N stores for all patches, which can then be used to initialize subsequent simulations.

- advanced method of spin-up to reach varying stages of observed ecological states
- run to reach spatially explicit targets, i.e. LAI
- types of spin-up targets
- sources for deriving targets

19.1 Prerequisites to running:

- spin-up target file
- target file header
- spin-up definition file
- worldfile with spinup object ID
- worldfile header pointer

19.2 Running a target driven spin up

- required command line option
- automatic world state generated (no tec event needed)
- setting the end date

20 Parallelization (The Accelerator)

- run a large watershed in parallel in order to speed up the simulation process
- compiling with openmp

Appendices (The Vault)

Command-line functionality

History/Legacy tools/functionality

Code organization

additional items will become clear as I go along!